

REPLICATOR NODE MODEL

The Replicator Node Model defines a class of nodes capable of creating exact, operationally-ready copies of themselves to expand Huey's computational footprint on demand. These nodes operate within strict governance limits to prevent uncontrolled proliferation.

Technical Specifications:

- Compute Core – Virtualized CPU instance with dynamically allocated RAM matching workload needs.
- Storage – Snapshot-based replication to ensure state consistency.
- Networking – Secure handshake protocol with orchestration layer before activation.

Governance Integration:

- All replication events are recorded in the Unified Audit Log (UAL) including initiator, target, timestamp, and resource cost analysis.
- Governance approval is required for replication beyond preset thresholds or outside strategic scope.

Operational Advantages:

- Rapid scaling during high-demand scenarios.
- Fault tolerance through redundancy.
- Task specialization by creating optimized replicas for specific workloads.

KPIs:

- Replication completion time (target ≤ 3 min).
- Error-free replication rate (%).
- Successful rollback rate after replication reversal.

Testing Objectives:

- Simulate over-replication event to trigger governance intervention.
- Validate orchestration load balancing post-replication.
- Measure system stability under continuous replication cycles.

Cross-References:

- See 'Node Types' for classification context.
- See 'Orchestration' for deployment workflow details.

[DIAGRAM PLACEHOLDER: 'Replicator Lifecycle and Governance Oversight Flow' — showing request, governance review, replication execution, and orchestration integration steps.]